

4. Professional Duties in Engineering

Introduction

Professional duties are the responsibilities that engineers must follow while doing engineering work. Engineering is not just a job; it is a profession because engineers use specialized knowledge, judgment, and skill for the benefit of society.

According to Fleddermann, codes of ethics explain the **rights, duties, and obligations** of engineers. These codes guide engineers in difficult situations, but they do not replace personal judgment. Engineers must use both professional codes and moral reasoning to make ethical decisions.

1. Duty to Protect Public Safety, Health, and Welfare

The most important duty of an engineer is to protect the safety, health, and welfare of the public. No duty is greater than this. Fleddermann explains that engineering codes strongly stress public safety because engineering products, structures, and systems directly affect human life.

An engineer must not approve unsafe designs, false test results, or weak safety procedures. If management pressures an engineer to ignore safety, the engineer should raise the issue and report it to the proper authority if needed.

Example:

If a bridge design is unsafe, the engineer must not approve it just because the company wants to save money.

2. Duty to Work Only in Areas of Competence

Engineers should accept only those tasks for which they are properly qualified by education, training, or experience. This duty protects the public from poor technical decisions.

If an engineer is not an expert in a certain area, they should take help from qualified specialists. Fleddermann's included codes also state that engineers should perform services only in areas of their competence.

Example:

A software engineer should not independently certify a civil structure unless they have proper civil engineering expertise.

3. Duty of Honesty and Truthfulness

Engineers must be honest in reports, designs, data, public statements, and professional opinions. They should not hide facts, exaggerate results, or mislead clients and the public.

The NSPE code in Fleddermann states that engineers should issue public statements only in an objective and truthful manner. This means opinions should be based on facts, knowledge, and competence.

Example:

If testing shows a product has failed, the engineer must report the failure honestly instead of changing the results.

4. Duty of Confidentiality

Engineers often handle secret or private information belonging to employers or clients. This may include designs, formulas, test results, software code, product plans, suppliers, costs, or business strategies.

According to Fleddermann, confidentiality is an important professional responsibility. Engineers must protect proprietary information unless disclosure is allowed by law or required to prevent serious harm.

Example:

An engineer should not take source code from a previous company and use it in a new company's product without permission.

5. Duty to Avoid Conflict of Interest

A conflict of interest occurs when an engineer's personal interest affects, or appears to affect, professional judgment. Engineers must act as faithful agents or trustees for their employer or client.

If a conflict exists, the engineer should disclose it clearly. Fleddermann explains that professional codes forbid conflicts of interest and advise engineers to use ethical judgment when the situation is unclear.

Example:

An engineer should not secretly recommend a supplier if they are receiving personal commission from that supplier.

6. Duty to Respect Intellectual Property

Engineers must respect the intellectual property rights of others. This includes patents, copyrights, designs, technical drawings, software, and confidential research.

The ASME code in Fleddermann says engineers should respect proprietary information and intellectual property rights. This duty is important because engineering work often depends on original designs, data, and technical ideas.

Example:

Using copyrighted software code from an old employer without permission is unethical, even if the engineer originally helped write it.

7. Duty of Fairness and Professional Conduct

Engineers should be fair, honest, and respectful toward clients, employers, colleagues, competitors, and the public. They should not use fraud, bribery, deception, false criticism, or unfair competition.

In competitive bidding, engineers must not submit false qualifications, lowball bids, or make dishonest claims just to win a project. Fleddermann explains that competitive bidding can create ethical temptations, so engineers must remain fair and honest.

Example:

An engineer should not win a contract by lying about experience or by making false claims against another engineer.

8. Duty of Professional Development

Engineering knowledge changes with time. Therefore, engineers should continue learning throughout their careers.

They should read technical literature, attend training, join professional activities, and improve their skills. Fleddermann's codes mention that engineers should maintain and improve their competence and support the development of other engineers.

Example:

A computer engineer should keep learning about cybersecurity, AI ethics, and data privacy because technology keeps changing.

9. Duty Toward Environment and Sustainable Development

Engineers should also consider environmental impact in their professional work. Modern codes include sustainability because engineering projects affect natural resources, pollution, energy use, and future generations.

The ASME and ASCE codes in Fleddermann include environmental impact and sustainable development as professional duties. This means engineers should not focus only on cost and profit.

Example:

An engineer should recommend a cleaner waste management system if the cheaper option will damage nearby land or water.

Memorization Checklist

Remember professional duties with:

S-C-H-C-I-F-D-E

S — Safety: Protect public safety, health, and welfare.

C — Competence: Work only in areas of qualification.

H — Honesty: Give truthful reports and statements.

C — Confidentiality: Protect client/employer information.

I — Intellectual Property: Respect copyrights, patents, and designs.

F — Faithful Agent: Avoid conflicts of interest.

D — Development: Keep improving professional knowledge.

E — Environment: Consider sustainability and environmental effects.

How to Apply in Case Studies

In a case study, first identify the engineer, employer, client, public, and other affected people. Then ask: **Which professional duty is being violated?**

Usually, the case will involve one of these conflicts:

- safety vs. company profit
- honesty vs. pressure from boss
- confidentiality vs. public danger
- personal benefit vs. client interest
- competence vs. accepting work for money

After that, apply the code of ethics. Write that the engineer must protect public welfare, act honestly, work within competence, avoid conflicts, and report serious danger if needed.

Short Case Study: Unsafe Testing

Suppose Robert is asked by his CEO to skip required safety testing and sign false documents. This violates professional duties because the engineer must protect public safety and must not falsify reports.

Robert should refuse to sign the false paperwork. He should explain the danger to the CEO and document his concern. If the company still forces him and public safety is at risk, he should report the matter to the proper authority.

This is because an engineer's duty to the public is higher than loyalty to the employer.

5-Mark Exam Answer

Professional duties are the responsibilities that engineers must follow while performing engineering work. According to Fleddermann, engineering codes of ethics explain the duties and obligations of engineers and guide them in ethical decision-making. The most important duty is to protect the safety, health, and welfare of the public. Engineers must not approve unsafe designs, false reports, or incomplete testing.

Engineers must also work only in areas of their competence. They should be honest and objective in reports, statements, and technical opinions. They must protect confidential and proprietary information of employers and clients, but this duty does not allow them to hide serious danger to the public. Engineers should avoid conflicts of interest and act as faithful agents for their clients and employers.

They must also respect intellectual property, avoid deceptive acts, maintain professional development, and consider environmental impact and sustainability. Therefore, a good engineer

should balance loyalty to employer with a higher duty to public safety, honesty, professional competence, and responsible engineering practice.